

Subject Description Form

Subject Code	EEE514
Subject Title	Quantum Communication
Credit Value	3
Level	5
Pre-requisite / Co-requisite/ Exclusion	Nil
Objectives	This subject aims to provide students with knowledge and principles of quantum communication, an understanding of different quantum communication protocols, the study of quantum repeaters, quantum networks, and quantum photonic chips.
Intended Learning Outcomes	Upon completion of the subject, students will be able to: - Overview different quantum communication protocols, including teleportation, single-photon quantum key distribution (BB84), and entanglement-based quantum key distribution (E91). - Understanding of quantum communication technologies, including quantum repeaters, mode dispersion, attenuation, and error detection. - Explain fundamental concepts of quantum entanglement communication and quantum secure communication. - Explain measurements in quantum communication systems, such as Bell state measurements and stationary/flying qubits. - Synthesize the acquired knowledge to understand system-level integration of quantum communication, both quantum and classical, and secure communication and distributed quantum computation.
Subject Synopsis/ Indicative Syllabus	Basic concepts of quantum communications: Quantum states, unitary operations, measurement, quantum entanglement, Bell states, qubits, pure and mixed states, etc. Fundamentals of optics: Coherent light, single photons, constructive and destructive interference, total internal reflection, waveguides, optical fibers, etc. Quantum key distribution (QKD): Cryptographically secure communication. Private vs public key. BB84 Protocol (single-photon QKD). Eavesdropper detection. QKD chips and systems. Entanglement-based QKD. Quantum teleportation: Teleportation protocol. Entangled pair. Joint measurement. No-cloning theorem. Fidelity. Long-Distance quantum communication: Mode dispersion, attenuation, overcoming losses, and quantum challenges. Quantum repeaters: Link-level entanglement. Entanglement swapping. Error detection and purification. Quantum network.

Teaching/Learning Methodology	Lecture: The fundamentals of quantum communication and principles of various quantum communication protocols will be described. Students are free to request help. Students are encouraged to solve problems and to use their knowledge to verify their solutions before seeking assistance. Tutorial: A set of problems and group discussion topics will be arranged in the tutorial classes. Students are encouraged to solve problems before having solutions.																																							
Assessment Methods in Alignment with Intended Learning Outcomes	<table><tr><th rowspan="2">Specific assessment methods/tasks</th><th rowspan="2">% weighting</th><th colspan="5">Intended subject learning outcomes to be assessed (Please tick as appropriate)</th></tr><tr><th>a</th><th>b</th><th>c</th><th>d</th><th>e</th></tr><tr><td>1. Assignment</td><td>40%</td><td>✓</td><td>✓</td><td>✓</td><td>✓</td><td>✓</td></tr><tr><td>2. Examination</td><td>60%</td><td>✓</td><td>✓</td><td>✓</td><td>✓</td><td>✓</td></tr><tr><td>Total</td><td>100 %</td><td colspan="5"></td></tr></table> Continuous assessments consist of assignments and mid-term test. These will assess the students’ understanding of basic concepts and principles in quantum communication technologies. The examination will be conducted to make a comprehensive assessment of students’ intended learning outcomes as stated above.							Specific assessment methods/tasks	% weighting	Intended subject learning outcomes to be assessed (Please tick as appropriate)					a	b	c	d	e	1. Assignment	40%	✓	✓	✓	✓	✓	2. Examination	60%	✓	✓	✓	✓	✓	Total	100 %					
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Student Study Effort Expected	<table><tr><td>Class contact:</td><td></td></tr><tr><td>▪ Lecture</td><td>33 Hrs</td></tr><tr><td>▪ Tutorial</td><td>6 Hrs</td></tr><tr><td>Other student study efforts:</td><td></td></tr><tr><td>▪ Reading and self-study</td><td>81 Hrs</td></tr><tr><td>The total student study effort</td><td>120 Hrs</td></tr></table>						Class contact:		▪ Lecture	33 Hrs	▪ Tutorial	6 Hrs	Other student study efforts:		▪ Reading and self-study	81 Hrs	The total student study effort	120 Hrs																						
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Reading List and References	Textbook - Gianfranco Cariolaro, Quantum Communications, Springer Cham, 2015. References - Ivan B. Djordjevic, Quantum Communication, Quantum Networks, and Quantum Sensing, Academic Press, 2022. - Mohsen Razavi, An Introduction to Quantum Communications Networks, Morgan & Claypool Publishers, 2018. - Vinod K. Mishra, An Introduction to Quantum Communication, Momentum Press, 2016.																																							